Summative Assessment — Submission Document

AI in Action: Building Intelligent Vision Systems for Smart Cities & Smart Retail

Scenario 1 — ParkVision AI: Intelligent Urban Parking Analytics & Space Optimisation Platform

1 · Project links

GitHub repository	FILL_IN — https://github.com/<username>/<repository>
Live Streamlit app	FILL_IN — e.g. https://parkvision-ai.streamlit.app
Assessor access granted to	ai.assignments@wacpinternational.org

Both links are clickable. The repository contains all code, the trained model, the processed dataset, the evaluation results and a detailed README.

2 · Candidate details

Student's full name	**FILL_IN — your full name**
Candidate registration number	**FILL_IN — your candidate registration number**
CRS name	Artificial Intelligence
Course name	Machine Learning and Deep Learning
School name	**FILL_IN — your school name**
Type of assessment	Summative Assessment (Individual)
Date of submission	07 September 2026

3 · Project summary

ParkVision AI analyses a photograph of a parking lot and reports, bay by bay, which spaces are occupied and which are free. Every bay is perspective-corrected out of the photograph and classified by a MobileNetV2 network fine-tuned on the PKLot dataset. The per-bay results are aggregated into availability counts, an occupancy percentage, a congestion level and a recommendation, then presented through a Streamlit dashboard with colour-coded overlays — green for empty, red for occupied, amber where the model was unsure.

Approach	Slot-level image classification over surveyed bay geometry
Model	MobileNetV2 (ImageNet), two-phase transfer learning
Dataset	PKLot — Almeida et al., Expert Systems with Applications, 2015
Interface	Streamlit dashboard, deployed on Streamlit Community Cloud

4 · Dataset

Split	Empty	Occupied	Total
Train	2,100	2,100	4,200
Val	450	450	900
Test	450	450	900
Total	**3,000**	**3,000**	**6,000**

Built from 12,142 PKLot photographs across 3 parking lots and 3 weather conditions, resized to 224x224 pixels. **Split strategy:** whole (lot, calendar date) groups are assigned to a single split, so no day appears in more than one split and the reported accuracy is not inflated by near-identical frames taken five minutes apart.

5 · Results

Metric	Value
Test accuracy	99.89%
Precision (occupied)	99.78%
Recall (occupied)	100.00%
F1 score (occupied)	99.89%
ROC AUC	1.0000
Errors on the held-out split	1 of 900 bays
Training	22 epochs (12 head + 10 fine-tune)

Accuracy by weather condition

Weather	Bays tested	Accuracy
Cloudy	222	99.55%
Rainy	157	100.00%
Sunny	521	100.00%

End-to-end testing on 12 held-out whole-lot photographs: mean slot accuracy 99.92%, mean error in the free-bay count 0.08 bays, congestion level called correctly on 12 of 12 scenes.

6 · Repository checklist

- All project code and script files (**.py** and **.ipynb**)
- Functional link to the deployed Streamlit web application
- The dataset used in the project (**data/processed/**, 6,000 labelled slot crops)
- The trained model (**models/parkvision_mobilenetv2.keras**)
- Detailed **README.md** covering research findings, references, data preparation, models and training parameters, accuracy metrics, and screenshots of the demo
- Repository accessible to **ai.assignments@wacpinternational.org**

Before submitting: the fields shown in amber above are still placeholders. Fill them in **submission/student_details.json** and re-run **python scripts/05_make_submission.py** to regenerate this PDF. Outstanding: student_full_name, candidate_registration_number, school_name, github_username, github_repository, streamlit_app_url